GRC Transactions, Vol. 42, 2018

Using Principal Component Analysis to Aid in Visualization and Interpretation of Geothermal Solute Chemistry: An Application to Yellowstone Thermal Waters

Jon K. Golla

University of New Mexico

Keywords

Geochemistry, Multivariate Statistics, Principal Component Analysis, Yellowstone

ABSTRACT

Principal Component Analysis (PCA) is a multivariate statistical technique, which aims to summarize data by efficiently displaying the common patterns (natural causes) of variation in lower-dimensional plots. Although widely used in topics such as lower-temperature aqueous geochemistry, ecology, etc., PCA has not garnered much utilization in geothermal science. This technique is applied to 100 select water chemistry samples from Yellowstone National Park literature where a complete data suite (major chemistry, trace metals, rare earth, etc.) is available. Preliminary results feature reaffirmed interpretation of 8 parameters of major chemistry. This investigation reveals proper classification of waters into alkali-chloride (Na+K-, Cl-, and HCO_3-rich and higher pH), acid-sulfate (low pH and SO_4-rich), shallow dilute (Ca+Mg-rich), and deep geothermal (SiO_2- and high-temperature-rich) groupings. Samples that plot in between show affinity for two or more clusters. A second PCA application reveals associations of trace element chemistry with these previously defined hydrogeochemical classes. High Pb, Ba, Zn, V, Fe, Al, Th, Y, Eh, ∑REEs content appear strongly correlated to each other and to acid-sulfate springs. Comparatively, there may be a link between As, B, Sb, Li, Cs, Mo, W and alkali-chloride waters. This paper focuses on coupling multivariate statistics and solute geochemistry to ease visualization and interpretation of observed hydrogeochemical variations from a well-studied geothermal system. Among many other objectives, this work also aims to provide a careful outline of methodology and benefits of the application of PCA to geothermal geochemistry data.

1. Introduction

Exploratory data analysis through multivariate statistics aims to simplify interpretation of large data sets by revealing any latent structures. These techniques have been widely used in various topics such as psychology (Stevens, 2012), signal processing (Scholkopf et al., 1997), ecology (James and McCulloch, 1990), and, more closely related, low-temperature aqueous geochemistry (Chen et al., 2007; Cloutier et al., 2008; Guler et al., 2002; Olsen et al., 2012; Sergeant et al., 2016). However, despite some exceptions (Ajima et al., 2010; Fowler et al., 2017; Li et al., 2011; Lindsey et al., 2018; Salvania and Nicholson, 1990; Veldeman et al., 1990), multivariate statistics have not garnered much utilization in the geothermal sciences. This study focuses on application of Principal Component Analysis or PCA (Hotelling, 1930; Pearson 1900), arguably the most commonly used multivariate statistical technique (Abdi and Williams, 2010; Krzanowsi, 1987), as a supplementary tool in visualization and interpretation of geothermal solute chemistry. PCA allows for dimensionality reduction or data compression by maximizing variance of standardized data along lower dimensions to determine underlying patterns (or, in this case, natural processes) responsible for variation. All data analyses and investigations are executed in R, a popular computing environment for statistical programming (R Core Team, 2018). Custom packages from the Comprehensive R Archive Network or CRAN are employed to aesthetically enhance plots and streamline statistical implementations (Hamilton, 2017; Kassambara and Mundt, 2017; Oksanen et al., 2018; Sarkar, 2008).

2. Yellowstone Water Chemistry

The well-studied geothermal system in the 0.64-Ma Yellowstone Caldera within Yellowstone National Park is a surface expression of a ~16-Ma volcanic hotspot, which is believed to have initiated its northeastern track across the North American tectonic plate in southeastern Oregon and northern Nevada where McDermitt Caldera stands (Christiansen, 1984; Pierce et al. 1992). Early geochemical investigations of Yellowstone thermal waters began in the late 1800s (Allen and Day 1935; Gooch and Whitfield, 1888). Subsequent studies later proposed that these waters may be derived from a single high-temperature (~335-340 ^{o}C), meteorically recharged deep water, while also suggesting that any significant differences in spring chemistry at the surface are a result of variable boiling and groundwater mixing (Fournier, 1989; Rye and Truesdell, 2007; Truesdell et al., 1977). Gas chemistry reflects interpretations correspondent to hydrogeochemistry, delineating shallow crustal and deep mantle end-members (Chiodini et al., 2012; Kennedy et al., 1985, 1987).

This study uses an extensive thermal water chemistry database (major chemistry, trace metals, rare earth elements, etc.) compiled by McCleskey et al. (2014) from 19 Yellowstone thermal areas, including all major geyser basins on the western side of the caldera. More details regarding sampling locations and analytical protocols are contained in accompanying report documents. The original data set of 122 samples is reduced to 100 samples; details about this sample truncation is further discussed in this section and in subsection 3.1.

The major chemistry of this data set is summarized with four hydrogeochemical groups, mainly by the classical Cl-SO_4-HCO_3 ternary diagram (Figure 1a). The Yellowstone geothermal waters show characteristics that are typical of surface manifestations commonly found in volcanogenic

geothermal environments. Most of these thermal features have retained constant chemical compositions, except for some thermal waters within Norris Geyser Basin, which have notably been observed to drastically change chemistry (Fournier et al., 2002; Nordstrom et al., 2009; White et al., 1988). Generally, there are two prominent facies (acid-sulfate and alkali-chloride), which reflect the end-members found in noted hot-water and vapor-dominated subsystems at Yellowstone (Fournier, 1989). There is a smaller set of relatively high-pH waters with considerable Cl content that shows an affinity to HCO_3, and, hence, these samples may be identified as alkali-bicarbonate-chloride waters. The rest of the waters have intermediate signatures. Color coding of figures of more descriptive chemistry and PCA results are based on these hydrogeochemical classes.

The Yellowstone thermal waters are sourced meteorically (Rye and Truesdell, 2007; Truesdell et al., 1977), and their varying chemical signatures may be traced to multiple subsurface processes with stable water isotopes (Figure 1b). Boiling and rock-water interactions are respectively responsible for end-member acid-sulfate and alkali-chloride compositions, while other intermediate chemical signatures may result from a combination of the former two processes or fluid mixing.

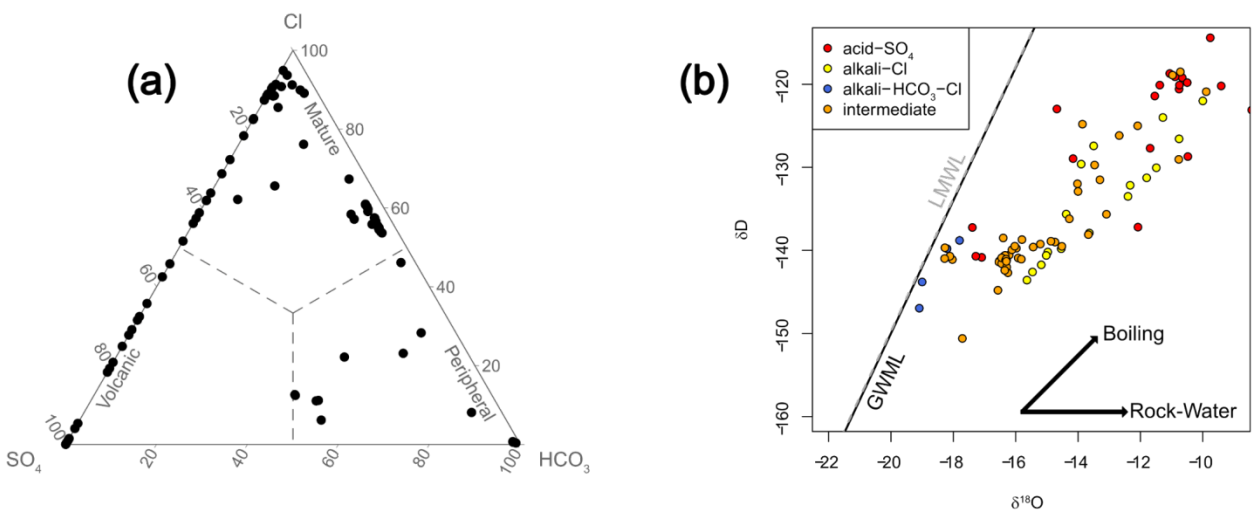

Figure 1: Major chemistry of Yellowstone thermal waters: (a) Cl-SO_4-HCO_3 ternary plot; (b) stable water isotope chemistry, with Global Meteoric Water Line (GWML) and Local Meteoric Water Line (LWML) respectively from Craig (1961) and Thordsen et al. (1992).

The distribution of trace metals is shown with a dot plot (Figure 2a). Trace metals measurements with ≥30% non-detects and/or below-detection values are removed and not shown. The range of relative abundance is defined by 'ultra-trace' elements (Ni, Pb, Sb, U, etc.) measuring as low as ~0.1-1 ppb and elements (B, Al, Fe, Li, etc.) approaching the lower range of natural water major-ion concentrations (~1-10 ppm). A discernable pattern is not easily recognized, as about 61% of the 23 retained trace metals appear the most concentrated in waters of intermediate compositions. Furthermore, the logarithmic scaling of this plot makes differentiating data points similar in relative concentration difficult, as samples are often interspersed and/or overlapping.

Analyzed rare earth element (REE) data show patterns congruent with those observed by Lewis et al. (1997) (Figure 2b). Generally in most waters, a 'see-saw' or 'gull-wing' pattern is typified by enrichment of light REEs (La to Gd), relative depletion of heavy REEs (Tb to Lu), and a pronounced negative Eu anomaly. This shared overall REE pattern represents a very similar signature shown by local Lava Creek tuff (Lewis et al., 1997). End-members may be more readily defined relative to total REE content. Acid-sulfate and acid-sulfate-chloride (belong to intermediate group; these orange REE patterns intermingle with the red ones) waters are the most concentrated. The alkali-chloride waters are relatively enriched. Some intermediate (below 10^{-4} normalized concentration) and alkali-bicarbonate-chloride waters appear below detection.

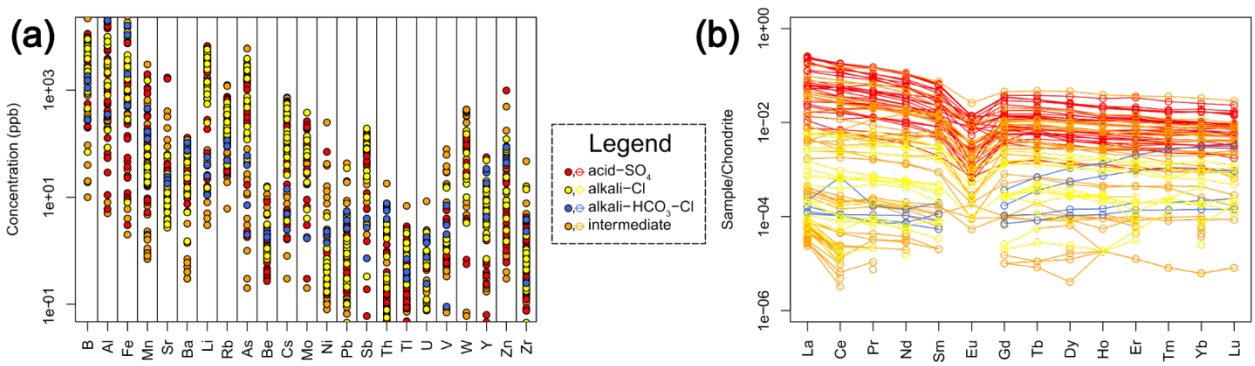

Figure 2: Trace element chemistry of Yellowstone waters: (a) dot plot of trace metal chemistry; (b) REE patterns – normalization after McDonough and Sun (1995).

3. Methodology

The first analysis is applied to major chemistry data, and an accompanying generalized code can be found in the Appendix. Below is a step-by-step outline of the general workflow:

3.1 Step 1: Prepare the data

The data are set up in an *m* x *n* matrix such that rows (*m*) are samples and columns (*n*) are the variables. PCA typically works best when *samples* >> *variables*. Ideally, the data matrix should be rectangular or continuous and devoid of any 'blanks'. There are imputation methods for missing values (Josse and Husson, 2016; Kim and Curry, 1977; Little and Rubun, 2014), but those are not explored here. For simplicity, samples with non-detect measurements are removed. Moreover, there are below-detection values in the working data set, and this occurrence is usually not uncommon in geochemical analyses. This study uses a method found in the USGS-RASS-STATPAC system (Güler et al., 2002; Lindsey, pers. comm.; VanTrump Jr. and Miesch, 1977) to replace below-detection concentrations with ¾ of the detection limit, but other substitution methods exist, like replacement with ½ of the detection limit via Farnham et al. (2002). To ensure multivariate normality, the data are transformed and scaled. The data set consists of solute geochemistry and physiochemical parameters. A logarithmic transformation (different from log-ratio transformation used in compositional PCA – more details later in Section 5) is preferred to account for different measurement scales among aqueous chemistry

parameters (^{o}C vs ppm vs ppb vs pH vs etc.). Then, replaced and log-transformed columns are centered to zero mean and normalized to one standard deviation.

3.2 Step 2: Perform PCA

This study is directed towards the general practitioner, but the reader is referred to cited references in this subsection and especially to Jolliffe (2002) for a more detailed mathematical basis or proof. Here, a generalized explanation of the PCA algorithm is presented.

PCA may be expressed as a linear transformation in the form

$$P = TD, \tag{1}$$

where a rotation matrix T is applied to remap the prepared data matrix D to a new feature space P (Abdi and Williams, 2010; Shlens et al., 2014). First, the T matrix is a set of principal components (PC), wherein each column or PC may be interpreted as a linear combination of 'loadings' (measure of correlation between original geochemical variables and corresponding PC). Then, the P or 'scores' matrix is the rotated observations, wherein each column is a linear combination of the standardized data with loadings as the coefficients. There are two algorithms to derive T and P.

The first method, used by '`princomp()`' in R, is a spectral decomposition of the covariance (or, sometimes, the correlation) matrix of D (R Core Team, 2018; Shlens, 2014). The eigenvectors of the covariance matrix are the PCs. The sum of the eigenvalues equals the total variance. Hence, the PCs are organized in a matrix from greatest to least eigenvalue, so that the leading PCs account for most of the variation in the data. With reference to Equation 1, this arranged PC matrix is T and is multiplied to D to obtain P.

The second method, automated by the '`prcomp()`' function, is a singular-value decomposition or SVD (Eckart and Young, 1936) of the D matrix (R Core Team, 2018; Shlens, 2014; Wall et al., 2003). The SVD operation is defined by

$$D = U\Sigma V^T \tag{2}$$

where D, any rectangular m x n matrix (or, in this case, the data matrix), is factored into two orthonormal matrices, U (m x m) and V (n x n), and a diagonal m x n matrix, Σ. This method has similar output to the first because the elements of the decomposition are related to the elements of the covariance matrix of D (cov(D)). The columns of V are equivalent to the eigenvectors of cov(D). The square of Σ is proportional to the eigenvalues of cov(D). The scores or P from Equation 1 are given by the product of V and D. Note that if the positions (row- vs column-assignment) of variables and samples are reversed, then the columns of U would correspond to the eigenvectors of cov(D).

The robustness of the second/SVD method has been noted to be computationally more stable and efficient (Chambers, 1977; Gnanadesikan, 1977; Praus, 2005). Therefore, the '`prcomp()`' function is favored here.

3.3 Step 3: Determine appropriate number of principal components

Many rules-of-thumb exist in selecting the appropriate number of principal components to use, and these methods are thoroughly examined in Jackson (1993) and Jolliffe (2002). Three of those criteria are briefly discussed here, given their straightforwardness and convenience (thanks to R implementations). A simple approach is to choose PCs that account for some ideal or desired cumulative percentage of total variation. Generally, a good cutoff is within 70-90% (Jolliffe,

2002) -- this threshold is initially crossed with just the first two PCs, which account for ~73% of variance (Figure 3a).

The latter stopping methods are observed on a PC-versus-eigenvalue plot, commonly referred to as a scree plot (Figure 3b). Kaiser's rule (Kaiser, 1960) only retains PCs with variances or eigenvalues that exceed 1. According to this rule, two or three PCs may be retained for interpretation (the third PC is slightly just above the eigenvalue cutoff). Alternatively, the broken-stick model, a null model that generates a random distribution of eigenvalues, may be used comparatively (Frontier, 1976). A 'stick' of unit length is broken up into the same number of PCs and is sorted from longest to shortest. This process is repeated many times, so that the generated distributions of all permutations are averaged into one model, which is shown against real observations on the scree plot. Then, only PCs with eigenvalues that exceed the values generated by the model are interpreted. Accordingly, only the first two PCs pass.

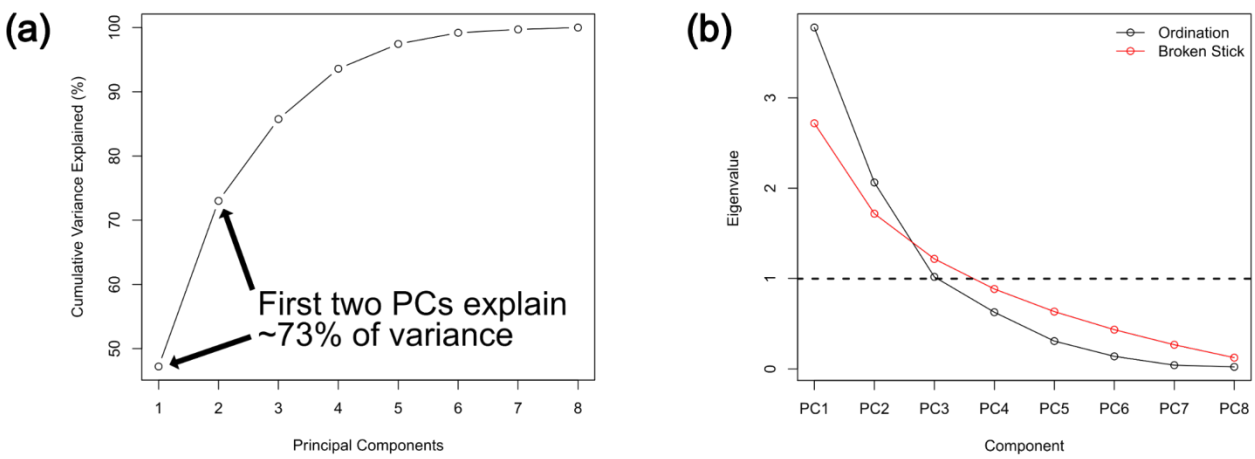

Figure 3: A (a) cumulative variance plot and (b) a scree plot with broken-stick model and Kaiser rule overlays.

The first two stopping rules have been noted to often yield an unnecessary (excessive) number of retained PCs (Jackson, 1993; Jollife, 2002). There is associated arbitrariness, especially with the cumulative variance rule because there is no strict standard to follow, so the broken-stick model is preferred. Many investigators (Jackson, 1993; Jollife, 2002; Legendre and Legendre, 1998) deem the broken-stick model as one of the more accurate approaches. Olsen et al. (2012) recently confirmed the model's reliability with extensive sensitivity analyses on water quality data from the Illinois River.

3.4 Step 4: Examine loadings

Ultimately, the decision to retain the appropriate number of PCs is still up to the practitioner. The interpretation from a PC must make sense, so examining loadings for a PC may be helpful after application of a stopping criterion. The broken-stick model had previously ruled in only two PCs, but the first three PCs are observed for demonstration (Figure 4).

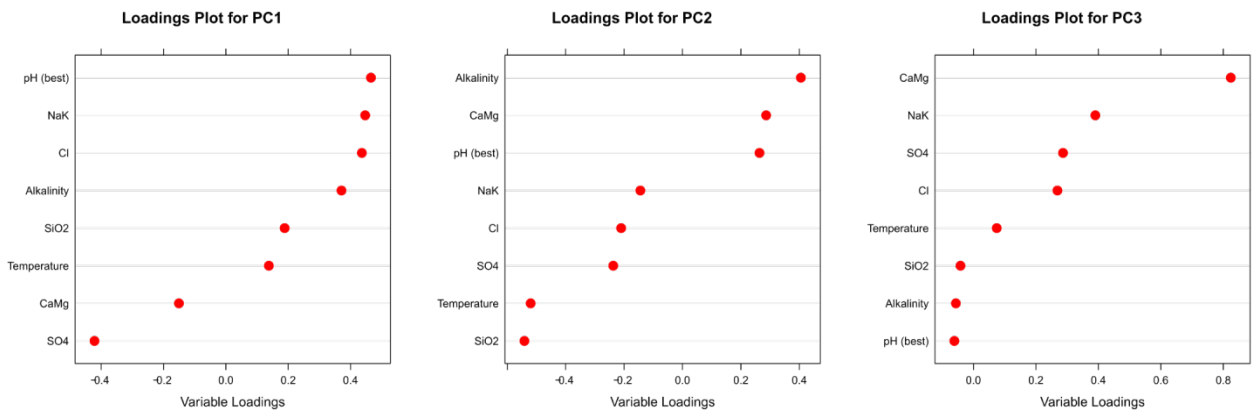

Figure 4: Dot plots of loadings for the first three components of the major-ion chemistry PCA application.

The square of a loading is a measure of correlation to the corresponding PC, so significantly high and low loadings are considered to be important to a given process or source. For PC1 and PC2, the most important loadings are (1) SO_4, Cl, Na+K, and pH and (2) Alkalinity, Ca+Mg (to an extent) Temperature, and SiO_2, respectively. PC1 may be attributed to a contrast between acid-sulfate (high SO_4) and alkali-chloride (high Cl, Na+K, pH) waters. PC2 may be associated to marked differences between shallow groundwater mixed (increased HCO_3 and Ca+Mg) and deep geothermal (hotter, increased SiO_2 content) end-members. Keeping just two PCs is reasonable, since variations observed from earlier descriptive major-ion chemistry are accounted for by interpretations of these components. Besides Ca+Mg, no other significant loadings contribute to PC3. Hence, this PC may yield unnecessary or redundant (shows strong association with Ca+Mg, like PC3) information.

3.5 Step 5: Plot and interpret results

Given that the aim of PCA is dimensionality reduction, it is best to reduce the number of PCs that characterize most of the variance of the original data set minimally (≥2 dimensions). An optimal way to plot PCA results with respect to the first two components is through the biplot (Gabriel, 1971; Jolicouer and Mossiman, 1960). There is convenience in this utility, since the biplot space shows loadings and scores simultaneously, enabling observations of relationships and groupings easier. For PCA interpretations involving more than two PCs, multiple biplots may be used to examine subsets of PC pairs; a 3-D biplot may be employed for three PCs (Weiner, 2017).

Two biplots from the same analysis are presented (Figure 5). Loadings radiate out from the origin as vectors, whose lengths are proportional to their influence on variance. Lower angles between vectors signify more correlation, and vice-versa. Scores or samples that plot farther from the origin along the trend of a loading are more greatly associated to the corresponding original variable. The first biplot (Figure 5a) captures interpretations of PC1 and PC2 from the last step, while also correctly representing hydrogeochemical variation in Figure 1a with a more complete set of geochemical parameters. The second biplot (Figure 5b) features the same PCA results, but scores are now classified with a color gradient from the adiabatic quartz geothermometer (Fournier and Rowe, 1966). Since abundant information is already presented on a biplot, *Step 4* may be forgone, since it is arguably integrated in the visualization.

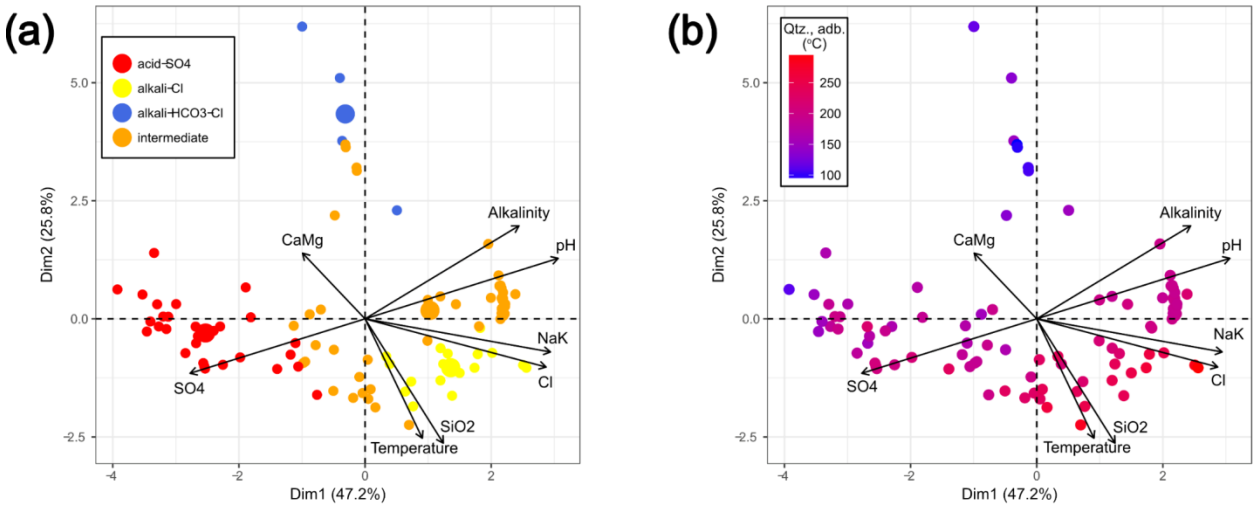

Figure 5: Biplot of major-ion chemistry with (a) color coding of hydrogeochemical grouping and (b) color gradient of the adiabatic quartz geothermometer (Fournier and Rowe, 1966).

4. Application to Trace Element Chemistry

The second PCA application (Figure 6) provides a clearer depiction of trace metal chemistry (Figure 2a). Preliminarily, traditional interpretation relative to a PC is less emphasized here, while relationships between variables and samples are more observed closely. Chemical affinities of hydrogeochemical groups are more obvious. Acid-sulfate waters appear strongly associated to high Pb, Ba, V, Fe, Th, Y, Zn, U, Mn, Be, Al, ∑REEs, Eh content. The preceding metals must favor low-pH, oxidizing conditions. Conversely, there seems to be a link between alkali-chloride waters and As, B, Sb, Li, Cs, Mo, W. Uniquely, the alkali-bicarbonate-chloride group shows correlation with Sr. One explanation for this coupling is groundwater interaction with carbonate or igneous rock, which has taken up Sr in exchange for Ca or K. Moreover, it is also much easier to see which end-members make up the composition of intermediate waters. To cross-validate these preliminary observations, a second PCA is iterated with trace element and major-ion chemistry (Figure 2b). The generated biplot is almost identical, and so is variance explained (~65% vs ~64%). At least for this instance, this check verifies that a PCA of the whole working data set yields complete interpretations of major-ion and trace element chemistry. In fact, more information may be presented. For example, there is now more statistical support to hypothesize that the Sr source may be more closely associated to carbonate dissolution than to interaction with igneous rock (Figure 6b).

5. Conclusions and Future Work

PCA is used to maximize variation of a multivariate data set to as few dimensions as possible. The ordination technique is applied to an extensive database of Yellowstone geothermal aqueous geochemistry, with a condensed 5-step workflow. The first PCA attempt successfully reaffirmed known and previously established major-ion chemistry groupings. The second iteration highlighted relationships of chemical parameters, elements, and samples, all of which are not always readily observable in a cluttered (logarithmic dot plot) or limited (ternary diagram) plot

or with just any one geochemical plot. This case study demonstrates that PCA is a useful *supplemental* tool for exploring patterns in a large geothermal hydrogeochemical data set.

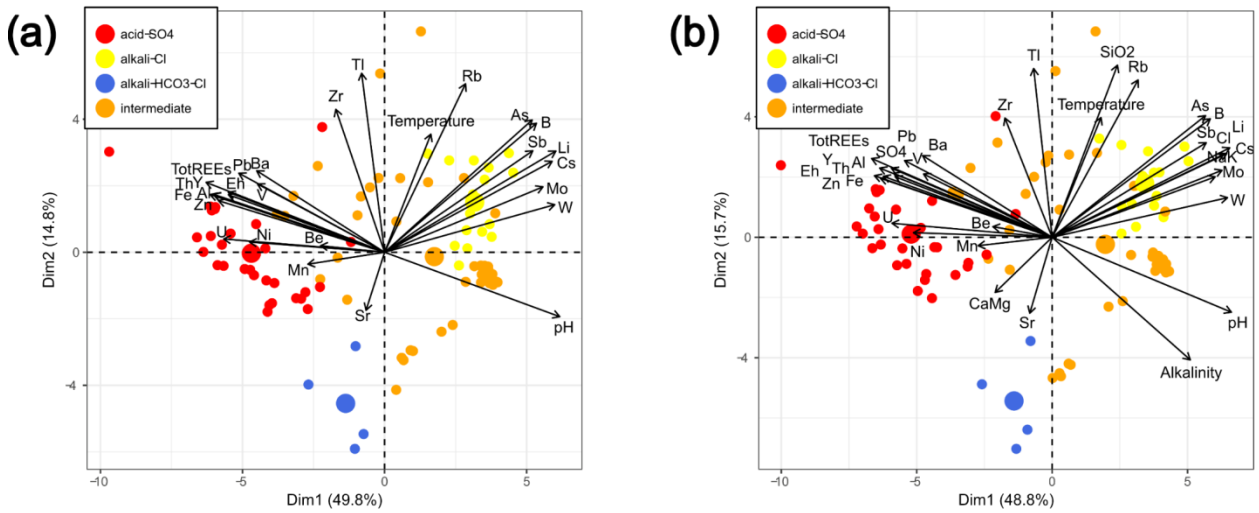

Figure 6: The following are biplots of (a) trace element chemistry and (b) trace element and major-ion chemistry. Broken-stick model outcomes for both iterations resulted in retaining two PCs.

However, conclusions of the analysis must be geologically possible and reasonable. Sole reliance on PCA and other statistical methods elicits risk of misinterpretation or partial interpretation (i.e., projection of geochemical data onto a new feature space disables the practitioner to infer from any obvious patterns based on actual concentrations and measured values). Moreover, only a non-compositional PCA is applied to the geochemical data set in this study; compositional PCA work (Aitchison and Greenacre, 2002; Filzmoser et al., 2009; Lindsey et al., 2018) is required for a further check on these non-compositional interpretations. Nonetheless, results of the pilot analyses are promising, and likely indicate the potential utility of PCA in other related applications. Interpretation of intermediate signatures may aid in water quality impact assessments (i.e., salinization, acidification, etc.) of volcanic geothermal systems on surrounding water resources. PCA may also be used for alternative hydrogeochemical mixing models, like M3, (Laaksoharju et al., 1999), which relies on a whole geochemical data set versus just one or a couple of tracers. Other future work may include:

- Performing sensitivity analyses by comparing results of compositional vs non-compositional (this paper) PCA
- Fingerprinting unknown resources with known resources
- Incorporating unexplored geochemical variables (stable water isotopes, gas geochemistry, etc.) into PCA
- Aligning PCA observations with more site-specific (versus generic) geochemical observations
- Investigating multidisciplinary PCA – combine aqueous geochemistry with other numerical data (from physical parameters, geophysics, hydrogeology, etc.)

6. Acknowledgements

I thank Cary Lindsey and Jerry Fairley of University of Idaho for providing relevant resources and inspiration towards use of multivariate statistics. Further discussion with Cary Lindsey helped improve the manuscript.

REFERENCES

Abdi, H. and Williams, L.J. "Principal component analysis." *Wiley interdisciplinary reviews: computational statistics* 2.4 (2010): 433-459.

Aitchison, J. and Greenacre, M. "Biplots of compositional data." *Journal of the Royal Statistical Society: Series C (Applied Statistics)* 51.4 (2002): 375-392.

Ajima, S., Todaka, N., and Akasaka, C. "Multivariate Analysis for Chemical Evolution of Reservoir Water for 35-year Operation of the Onikobe Geothermal Plant, Japan." *Proceedings of the World Geothermal Congress* (2010).

Allen, E. T., and A. L. Day. "Hot springs of the Yellowstone National Park. Carnegie Institute of Washington publication 466." (1935).

Chambers, J.M. “Computational Methods for Data Analysis.” *Wiley*, New York (1977).

Chen, K., Jiao, J.J., Huang, J., and Huang, J. "Multivariate statistical evaluation of trace elements in groundwater in a coastal area in Shenzhen, China." *Environmental Pollution* 147.3 (2007): 771-780.

Chiodini, G., Caliro, S., Lowenstern, J.B., Evans, W.C., Bergfeld, D., Tassi, F., and Tedesco, D. "Insights from fumarole gas geochemistry on the origin of hydrothermal fluids on the Yellowstone Plateau." *Geochimica et Cosmochimica Acta* 89 (2012): 265-278.

Christiansen, R.L. "Yellowstone magmatic evolution: Its bearing on understanding large-volume explosive volcanism." *Explosive Volcanism: Inception, Evolution, and Hazards* (1984).

Cloutier, V., Lefebvre, R., Therrien, R., and Savard, M.M. "Multivariate statistical analysis of geochemical data as indicative of the hydrogeochemical evolution of groundwater in a sedimentary rock aquifer system." *Journal of Hydrology* 353.3-4 (2008): 294-313.

Eckart, C. and Young, G. "The approximation of one matrix by another of lower rank." *Psychometrika* 1.3 (1936): 211-218.

Farnham, I.M., Singh, A.K., Stetzenbach, K.J., and Johannesson, K.H. "Treatment of nondetects in multivariate analysis of groundwater geochemistry data." *Chemometrics and Intelligent Laboratory Systems* 60.1-2 (2002): 265-281.

Filzmoser, P., Hron, K., and Reimann, C. "Principal component analysis for compositional data with outliers." *Environmetrics: The official journal of the International Environmetrics Society* 20.6 (2009): 621-632.

Fournier, R.O. "Geochemistry and dynamics of the Yellowstone National Park hydrothermal system." *Annual Review of Earth and Planetary Sciences* 17.1 (1989): 13-53.

Fournier, R.O., Weltman, U., Counce, D., White, L.D., and Janik, C.J. “Results of weekly chemical and isotopic monitoring of selected springs in Norris Geyser Basin, Yellowstone National Park during June-September, 1995.” *United States Geological Survey Open-File Report* 344 (2002).

Fowler, A.P.G., Spycher, N., Zeirenberg, R.A., and Cantwell, C.A. "Identification of blind geothermal resources in Surprise Valley, CA, using publicly available groundwater well water quality data." *Applied Geochemistry* 80 (2017): 24-48.

Frontier, S. "Decrease of eigenvalues in principal component analysis-comparison with broken stick model." *Journal of Experimental Marine Biology and Ecology* 25.1 (1976): 67-75.

Gabriel, K.R. "The biplot graphic display of matrices with application to principal component analysis." *Biometrika*58.3 (1971): 453-467.

Gnanadesikan, R. “Methods for Statistical Data Analysis of Multivariate Observations.” *Wiley,* New York (1977).

Gooch, F.A. and Whitfield, J.E. “Analyses of waters of the Yellowstone National Park, with an account of the methods of analysis employed.” *United States Geological Survey Bulletin* 47 (1888).

Güler, C., Thyne, G.D., and McCray, J.E. "Evaluation of graphical and multivariate statistical methods for classification of water chemistry data." *Hydrogeology journal* 10.4 (2002): 455-474.

Hamilton, N. “ggtern: An Extension to ‘ggplot2’, for the creation of Ternary Diagrams.” *R package version 2.2.1* (2017). <https://CRAN.R-project.org/package=ggtern>

Hotelling, H. "Statistical Methods for Research Workers." (1930).

Jackson, D.A. "Stopping rules in principal components analysis: a comparison of heuristical and statistical approaches." *Ecology* 74.8 (1993): 2204-2214.

James, F.C. and McCulloch, C.E. "Multivariate analysis in ecology and systematics: panacea or Pandora's box?" *Annual review of Ecology and Systematics* 21.1 (1990): 129-166.

Jolicoeur, P. and Mosimann, J.E. "Size and shape variation in the painted turtle. A principal component analysis." *Growth* 24.4 (1960): 339-354.

Jolliffe, I.T. “Principal Component Analysis.” *Springer Science & Business Media* (2002).

Josse, J. and Husson, F. "missMDA: a package for handling missing values in multivariate data analysis." *Journal of Statistical Software* 70.1 (2016): 1-31.

Kassambara, A. and Mundt, F. “factoextra: Extract and Visualize the Results of Multivariate Data Analyses.” *R package version 1.0.5* (2017). <https://CRAN.R-project.org/package=factoextra>

Kaiser, H.F. "The application of electronic computers to factor analysis." *Educational and psychological measurement* 20.1 (1960): 141-151.

Kennedy, B.M., Lynch, M.A., Reynolds, J.H., and Smith, S.P. “Intensive sampling of nobel gases in fluids at Yellowstone; I, Early overview of the data; regional patterns.” *Geochimica Cosmochimica Acta* 49 (1985): 1251-1261.

Kennedy, B.M., Reynolds, J.H., and Smith, S.P. “Helium isotopes; Lower Geyser Basin, Yellowstone National Park.” *Journal of Geophysical Research* 92 (1987): 12477-12489.

Kim, J.-O. and Curry, J. "The treatment of missing data in multivariate analysis." *Sociological Methods & Research* 6.2 (1977): 215-240.

Krzanowski, W.J. "Selection of variables to preserve multivariate data structure, using principal components." *Applied Statistics* (1987): 22-33.

Laaksoharju, M., Skårman, C., and Skårman, E. "Multivariate mixing and mass balance (M3) calculations, a new tool for decoding hydrogeochemical information." *Applied Geochemistry* 14.7 (1999): 861-871.

Legendre, P. and Legendre, L. "Numerical ecology: second English edition." *Developments in environmental modelling* 20 (1998).

Li, T., Zhu, J., and Zhang, W. “Principal Component Analysis of Heat Transfer for Ground Heat Exchanger.” *Geothermal Resources Council Transactions* 35 (2011): 1121-1126.

Little, R.J.A. and Rubin, D.B. “Statistical analysis with missing data.” *John Wiley & Sons* 333 (2014).

Lindsey, C.R. Personal interview. 28 June 2018.

Lindsey, C.R., Neupane, G., Spycher, N., Fairley, J.P., Dobson, P., Wood, T., McLing, T., and Conrad, M. "Cluster analysis as a tool for evaluating the exploration potential of Known Geothermal Resource Areas." *Geothermics* 72 (2018): 358-370.

McCleskey, R.B., Chiu, R.B., Nordstrom, D.K., Campbell, K.M., Roth, D.A., Ball, J.W., and Plowman, T.I. “Water-Chemistry Data for Selected Springs, Geysers, and Streams in Yellowstone National Park, Wyoming, Beginning 2009.” *United States Geological Survey* (2014). <https://water.usgs.gov/nrp/yellowstone-water-chemistry/>

McDonough, W.F. and Sun, S.-S. "The composition of the Earth." *Chemical geology* 120.3-4 (1995): 223-253.

Nordstrom, D.K., McCleskey, R.B., and Ball, J.W. "Sulfur geochemistry of hydrothermal waters in Yellowstone National Park: IV Acid–sulfate waters." *Applied Geochemistry* 24.2 (2009): 191-207.

Okansen, J., Blanchet, F.G., Friendly, M., Kindt, R., Legendre, P., McGlinn, P.R., O’Hara, R.B., Simpson, G.L., Solymos, P., Stevens, M.H.H., Szoecs, E., and Wagner, H. “vegan: Community Ecology Package.” *R package version 2.5-2* (2018). <https://CRAN.R-project.org/package=vegan>

Olsen, R.L., Chappell, R.W., and Loftis, J.C. "Water quality sample collection, data treatment and results presentation for principal components analysis–literature review and Illinois River watershed case study." *Water research* 46.9 (2012): 3110-3122.

Pearson, K. "X. On the criterion that a given system of deviations from the probable in the case of a correlated system of variables is such that it can be reasonably supposed to have arisen from random sampling." *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science* 50.302 (1900): 157-175.

Pierce, K.L., Morgan, L.A. and Link, P.K. "The track of the Yellowstone hot spot: Volcanism, faulting, and uplift." *Regional geology of eastern Idaho and western Wyoming: Geological Society of America Memoir* 179.322 (1992): 1-53.

R Core Team. “R: A Language and Environment for Statistical Computing.” *R Foundation for Statistical Computing*, Vienna, Austria (2018).

Praus, P. "Water quality assessment using SVD-based principal component analysis of hydrological data." *Water SA*31.4 (2005): 417-422.

Rye, R.O. and Truesdell, A.H. "The question of recharge to the deep thermal reservoir underlying the geysers and hot springs of Yellowstone National Park." *United States Geological Survey Professional Paper* 1717 (2007).

Salvania, N. V., and K. Nicholson. "Chemometrics applied to the fluid chemistry of geothermal fields in the Taupo Volcanic Zone, New Zealand." *Proceedings 12th NZ Geothermal Workshop*, Auckland, New Zealand (1990).

Sarkar, D. “Lattice: Multivariate Data Visualization with R.” *Springer Science & Business Media* (2008). <http://lmdvr.r-forge.r-project.org>

Schölkopf, B., Smola, A., and Müller, K.-R. "Kernel principal component analysis." *International Conference on Artificial Neural Networks*. Springer, Berlin, Heidelberg (1997).

Sergeant, C. J., Starkey, E.N., Bartz, K.K., Wilson, M.H., and Meuter, F.J. "A practitioner’s guide for exploring water quality patterns using principal components analysis and Procrustes." *Environmental monitoring and assessment* 188.4 (2016): 249.

Shlens, J. "A tutorial on principal component analysis." *arXiv preprint arXiv:1404.1100* (2014).

Stevens, J. P. “Applied multivariate statistics for the social sciences.” *Routledge* (2012).

Truesdell, A.H., Nathenson, M., and Rye, R.O. "The effects of subsurface boiling and dilution on the isotopic compositions of Yellowstone thermal waters." *Journal of Geophysical Research* 82.26 (1977): 3694-3704.

VanTrump Jr., G. and Miesch, A.T. “The US Geological Survey RASS-STATPAC system for management and statistical reduction of geochemical data.” *Computers and Geosciences* 3.3 (1977): 475-488.

Veldeman, E. "Thermal waters from south Bulgaria: a multivariate approach for evaluation and interpretation of analytical data." *Geothermal Resources Council Transactions* 14 (1990): 1537-1543.

Wall, M.E., Rechtsteiner, A., and Rocha, L.M. "Singular value decomposition and principal component analysis." In *A practical approach to microarray data analysis*. Springer, Boston, Massachusetts (2003): 91-109.

Weiner, J. “pca3d: Three Dimensional PCA Plots.” *R package version 0.10* (2017). <https://CRAN.R-project.org/package=pca3d>

White, D.E., Hutcinson, R.A., and Keith, T. “The geology and remarkable thermal activity of Norris Geyser Basin, Yellowstone National Park, Wyoming.” *United States Geological Survey Professional Paper* 1456 (1988).

APPENDIX

An R starter script for PCA:

```
# Load Data via file path (indicate row index for row names, if desired;
# separator is set to comma delimiter)
data <- read.table("C:/filepath..", header=TRUE, sep = ",", row.names=1)
# or via clipboard
data <- read.table(file = "clipboard", sep = ",", header=TRUE, row.names=1)


# Display tabulated data:
head(data)

# Optional: combined columns (e.g., for hardness (Ca and Mg) or Na+K, etc.)
data$NaK <- data[, #] + data[, #]
data$CaMg <- data[, #] + data[, #]
head(data)

# Select data columns (variables) of interest; 1 is added to avoid undefined
# values from logarithm of zeroes
log.data <- log(data[,c(#,#,#,#,#,#)]+1)
head(log.data)

require("lattice") #for dot plots
require("vegan") #for broken-stick model

# Run PCA using 'prcomp' (based on SVD), with mean-centering and unit-scaling
pca <- prcomp(logdata, center = TRUE, scale. = TRUE)
summary(pca)
bstick(pca)
screeplot(pca, bstick = TRUE, type = "lines", main = "",ylab=”Eigenvalue”)

# Extract cumulative variance
vars <- apply(pca$x, 2, var)
proportions <- vars / sum(vars)
cumvar <- cumsum(proportions)*100
plot(cumvar,type="b", xlab ="Principal Components",ylab="Cumulative Variance Explained (%)")

# Extract loadings matrix
load    <- pca$rotation
# DotPlot PC1; the function 'order' sorts loadings
loadings1 <- load[order(load[, 1]), 1]
dotplot(loadings1, main="PC1 Loadings", xlab="Variable Loadings", cex=1.5, col="red")
# DotPlot PC2; the function 'order' sorts loadings
loadings2 <- load[order(load[, 2]), 2]
dotplot(loadings2, main="PC2 Loadings", xlab="Variable Loadings", cex=1.5, col="red")
# DotPlot PC3; the function 'order' sorts loadings
loadings3 <- load[order(load[, 3]), 3]
dotplot(loadings3, main="PC3 Loadings", xlab="Variable Loadings", cex=1.5, col="red")

# Generate the BiPlot; for a customizable biplot, use “factoextra” (Kassambara and Mundt, 2017)
biplot(pca, cex=c(1, 1), pc.biplot = TRUE, xlim=c(-2,2))
```